

AVINASH NAYAL

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RESEARCH SUMMARY

Ph.D. candidate in Mechanical Engineering with 5+ years of research experience in advanced materials synthesis, polymer composite development, and surface/interface characterization for thermal management and energy applications. Expertise in wet chemistry processing of polymer-matrix composites, nanoparticle dispersion and encapsulation, and structure-property optimization using DSC, TGA, Raman spectroscopy, SEM, and XRD. Proven track record in first-principles computational materials modeling (DFT, Quantum ESPRESSO), Python-driven data analysis and automation, and cross-functional collaboration across national laboratory and university research teams. Author of 8+ peer-reviewed publications in high-impact journals.

CORE COMPETENCIES

Synthesis & Processing: Polymer composite fabrication, Solution casting, Solvent processing (DMF, acetone), Nanoparticle dispersion, Expanded graphite intercalation, Phase change materials (PCM), Shear-alignment of polymer films, Wet chemistry

Characterization: DSC, TGA, Raman Spectroscopy, XRD, SEM, Laser Flash Analysis (LFA), Four-point probe, Thermal conductivity Measurement, UV-Vis, Calorimetry

Computational & Data: DFT (Quantum ESPRESSO, Phonopy), Python (NumPy, automation, visualization), MATLAB, ANSYS (Fluent, APDL), COMSOL, First-principles modeling, DOE, Statistical analysis

Soft Matter & Polymers: Polyetherimide (PEI), Epoxy nanocomposites, Paraffin wax composites, Graphene/GO fillers, Polymer-matrix thermal interface materials, Structure-property relationships

Lab Operations & Safety: SOP development, Experimental design, Sensor calibration (thermocouples, RTDs), NI-DAQ, Test fixture design, Lab safety protocols, Mentoring

Communication & Leadership: Technical writing (8+ publications), Cross-functional collaboration, Grant/proposal contributions, Conference presentations (APS, ACS), Peer mentoring, Stakeholder management

EDUCATION

Ph.D. in Mechanical Engineering | University of Oklahoma, Norman, OK Aug 2020 – Present

Advisor: Dr. Jivtesh Garg | Focus: Advanced materials for thermal management and energy storage

Relevant Coursework: Nanoscale Heat Transfer, Polymer Science, Computational Materials Science, Advanced Thermodynamics

M.S. in Mechanical Engineering | Arizona State University, Tempe, AZ Aug 2018 – Jul 2020

Thesis: Study of Solar Thermophotovoltaic (STPV) Energy Conversion with Selective Metafilm Coatings

B.Tech in Mechanical Engineering | Nirma University, Ahmedabad, India Aug 2013 – May 2017

RESEARCH & ENGINEERING EXPERIENCE

Graduate Research Intern | Oak Ridge National Laboratory | Oak Ridge, TN Jan 2025 – May 2025

Materials Science and Technology Division | Mentor: Dr. Lucas Lindsay

- Investigated phonon-mediated thermal transport in energy-relevant semiconductor materials using first-principles DFT simulations (Quantum ESPRESSO, Phonopy), advancing predictive models for materials design.
- Quantified the effect of mechanical strain on GaN thermal conductivity by performing phonon calculations on relaxed and c-axis-constrained crystal structures, informing high-power electronics packaging design.
- Developed a Python-based post-processing and visualization framework that standardized analysis routines and file structures, reducing data processing cycle time by 40% and enabling reproducible workflows.
- Contributed to a minimum thermal conductivity modeling framework combining phonon property calculations with empirical transport methods to evaluate thermal transport limits in solid-state systems.
- Collaborated cross-functionally with physicists, materials scientists, and computational engineers to deliver engineering-grade results under tight project timelines.

Graduate Research Assistant | University of Oklahoma | Norman, OK Jan 2023 – Present

Micro/Nano Heat Transfer Lab | Ph.D. Advisor: Dr. Jivtesh Garg

- Designed and commissioned a custom Couette-shear extrusion apparatus for polymer-film processing under controlled alignment conditions, enabling systematic study of structure-property relationships in thermally conductive composite films.
- Established repeatable SOPs for polymer film deposition, priming, purge sequences, and nozzle-to-mold alignment, ensuring process reproducibility and lab safety compliance across multiple experimental campaigns.
- Collaborated on comparative testing of air, liquid, composite phase change material (CPCM), and hybrid cooling methods for battery thermal management, achieving uniform temperature control ($\Delta T < 2^\circ\text{C}$).
- Built custom thermal test fixtures using SolidWorks; integrated thermocouples, RTDs, heaters, and NI-DAQ systems for controlled thermal evaluation of encapsulated and composite material samples.
- Standardized DSC/TGA characterization workflows to connect material-level thermal properties to system-scale performance, improving data repeatability and reducing characterization turnaround time.

Graduate Research Assistant | University of Oklahoma | Norman, OK Aug 2020 – Dec 2022

Micro/Nano Heat Transfer Lab | Polymer Composite & Nanoparticle Processing

- Synthesized expanded graphite (EG) via wet chemistry intercalation using H_2O_2 and other oxidizing agents; optimized reaction conditions to achieve up to 4600% enhancement in thermal conductivity of resulting polymer composites.

- Fabricated EG/polyetherimide (PEI) composites via solution casting, yielding 3070% thermal conductivity enhancement (7.3 W/m-K) and 19 orders-of-magnitude increase in electrical conductivity (969 S/m) for thermoelectric applications.
- Optimized graphene/epoxy thermal interface materials through solvent selection (DMF vs. acetone), achieving 44% higher thermal conductivity and improved nanoparticle dispersion uniformity verified by SEM and Raman spectroscopy.
- Performed first-principles thermal transport modeling of hexagonal BC₂P, computing a thermal conductivity of 162 W/m-K at room temperature, contributing to computational materials discovery efforts.
- Mentored 2 junior researchers on thermal testing protocols, DSC/TGA/Raman characterization workflows, and Python-based data analysis, improving data consistency across the research group.

Master's Thesis Researcher | Arizona State University | Tempe, AZ

May 2019 – Jul 2020

Nano-Engineered Thermal Radiation Group | Advisor: Dr. Liping Wang

- Developed a coupled radiation-conduction-electrical MATLAB model for solar thermophotovoltaic (STPV) systems, validated against experimental results with < 5% deviation across varying solar concentrations.
- Optimized multilayer absorber/emitter metafilm coatings using Particle Swarm Optimization and fabricated thin-film coatings via PECVD-sputtering, achieving high spectral selectivity (0.4–1.6 μm).
- Engineered a vacuum-based experimental STPV test setup with silica microsphere spacers to maintain a controlled 100 μm gap, demonstrating negligible conduction loss and quantifiable radiative efficiency gains.

Research Intern | Indian Institute of Technology Gandhinagar | India

Aug 2017 – Jul 2018

Mechanical Engineering Department | DRDO-funded fuel cell systems project

- Led testing and overhaul of a 5 kWe diesel-powered fuel-cell reformer system, designing and sourcing heat exchangers, DAQ, thermocouples, MFCs, and gas chromatograph components for system monitoring and safety.
- Developed scale-up plans from 5 kWe to 30 kWe capacity using ASPEN modeling, mass/energy balances, and component specification, delivering technical documentation and procurement timelines for defense sponsors.

SELECTED PUBLICATIONS

8+ peer-reviewed journal articles | 2 manuscripts in preparation | 2 conference presentations (APS, ACS)

1. Nayal, A., Garg, J. "Expanded Graphite/Paraffin Composite Phase Change Materials: Effect of Solvent and Ultrasonication on Thermal Properties." Oral talk, APS March Meeting, 2023.
2. Danayat, S., Nayal, A., et al. "Superior Enhancement in Thermal Conductivity of Epoxy/Graphene Nanocomposites Through Use of DMF." *MethodsX*, 11, 102319 (2023).
3. Tarannum, F., Danayat, S., Nayal, A., et al. "H₂O₂-Intercalated Expanded Graphite for Large Enhancement in Thermal Conductivity of PEI/Graphite Nanocomposites." *J. Mater. Res. Technol.*, 33, 1420–1435 (2024).
4. Danayat, S., Tasnim, Z., Nayal, A., et al. "Role of Four-Phonon Processes in Thermal Conductivity of 2D Materials and Polymer/Nanofiller Composites." *RSC Nanoscale*, 16, 13755–13783 (2024).
5. Annam, R.S., Danayat, S., Nayal, A., et al. "Thickness-Dependent Thermal Conductivity of SrTiO₃ Thin Films on Silicon Substrate." *J. Vac. Sci. Technol. A*, 42(2), 022707 (2024).

TECHNICAL SKILLS

Materials Characterization: Raman Spectroscopy, DSC, TGA, Laser Flash Analysis (LFA 467), XRD, SEM, Four-point probe, Thermal conductivity measurement, Calorimetry, 3D Scanning (Photogrammetry)

Processing & Fabrication: Polymer composite synthesis, Solution/solvent casting, Nanoparticle dispersion, Expanded graphite intercalation, Thin film deposition (PECVD-sputtering), Mold/test fixture design, 3D printing

Computational Tools: Python (NumPy, data automation, visualization), MATLAB, C++, Quantum ESPRESSO, Phonopy, ANSYS Fluent/APDL, COMSOL, Engineering Equation Solver (EES)

Design & CAD: SolidWorks (GD&T, DFM/DFA, BOM), Fusion 360, TINKERCAD

Data & Documentation: DOE, Statistical analysis, Technical writing, SOP development, MS Office Suite

AWARDS & FELLOWSHIPS

- Dr. Kent & Kayleen Thomas Scholarship, University of Oklahoma (2025)
- Exxon Mobile Milam Scholarship, University of Oklahoma (2023, 2024)
- Milton J. Gordon Memorial Scholarship, University of Oklahoma (2023)
- Gallogly College of Engineering Scholarship, University of Oklahoma (2023)
- Nettie Vincent Boggs Graduate Fellowship, University of Oklahoma (2022)
- Jim & Bee Close Engineering Scholarship, University of Oklahoma (2021, 2022)
- John E. Francis Scholarship, University of Oklahoma (2021)